Scientific Computing 2026

Sampling distributions, confidence intervals, and bootstrap

Uncertainty after fitting

Lecture 6

Five waiting times, one fitted rate

The observations from Chapter 5 were

$$1,\ 2,\ 3,\ 4,\ 5 \text{ minutes.}$$

The exponential model gave

$$\widehat{\lambda} = \frac{1}{\bar{t}} = \frac{1}{3} \text{ per minute.}$$

How much would that estimate change if we collected five new waits?

The parameter stays fixed when the sample changes

Imagine repeating the observation process under one true rate λ.

$$(T_1, \ldots, T_n) \quad \longmapsto \quad \widehat{\lambda} = \frac{n}{\sum_i T_i}.$$

Each new sample produces a new estimate. Their distribution describes the estimation error under that rate.

In practice, we have one dataset and do not know λ.

The quantity whose uncertainty matters

An interval must refer to a specific population quantity.

Data	Possible target
Waiting times	Arrival rate λ or mean wait $1/\lambda$
Repeated measurements	Population mean μ
Independent test examples	Accuracy p of a fixed classifier

The variation of individual outcomes and the uncertainty of an estimated population quantity answer different questions.

Confidence intervals

Definition: Confidence interval with level $1-\alpha$

A rule that takes a sample $X = (X_1, \ldots, X_n)$ and returns $[L(X), U(X)]$, satisfying

$$\mathbb{P}_\theta\{L(X) \le \theta \le U(X)\} \ge 1-\alpha \qquad \text{for every allowed } \theta.$$

The probability concerns repeated samples from P_θ.

An **exact** guarantee holds at the stated sample size under the model. An **approximate** interval can have coverage different from its nominal level.

Ninety-five percent coverage



Each of forty Gaussian samples ($n = 25$, known $\sigma = 2$) gives an interval. The fixed true mean is 10. Orange intervals miss it.

What the confidence level means

Before seeing the data, both endpoints are random. The interval construction covers the fixed parameter in at least 95% of repeated samples when it has a 95% guarantee.

After observing the data, the endpoints have numerical values. The realized interval either contains the parameter or misses it.

A 95% confidence level does not assign a posterior probability of 95% to the parameter lying in the realized interval.

Quantiles choose the central probability

Definition: Quantile

For a CDF F and $0 < p < 1$,

$$Q_p(F) = \inf\{x : F(x) \geq p\}.$$

For a continuous, strictly increasing CDF, $F(Q_p(F)) = p$.

Write z_p for the standard Gaussian quantile.

$$z_{0.975} \approx 1.96, \qquad \mathbb{P}(-1.96 \leq Z \leq 1.96) \approx 0.95.$$

A Gaussian mean with known variance

Assume $X_i \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, with known $\sigma > 0$.

$$Z = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \sim \mathcal{N}(0, 1).$$

The standardized error has a known law that does not depend on μ. Such a quantity is called a **pivot**.

$$\mathbb{P}_\mu \left\{ -z_{1-\alpha/2} \leq \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \leq z_{1-\alpha/2} \right\} = 1 - \alpha.$$

An interval by rearranging the inequalities

Result: Gaussian mean, known variance

Under the preceding Gaussian assumptions, an exact interval is

$$\left[\bar{X} - z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \bar{X} + z_{1-\alpha/2}\frac{\sigma}{\sqrt{n}}\right].$$

Larger σ produces a wider interval. Four times as many observations halves its width. Higher confidence increases the quantile and widens the interval.

The noise variance usually needs estimation

Use the sample variance from Chapter 5,

$$S^2 = \frac{1}{n-1}\sum_i (X_i - \bar{X})^2, \qquad n \geq 2.$$

Replacing σ by S also puts random variation into the denominator:

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S}.$$

An unusually small S can make this standardized error large. The Gaussian cutoff 1.96 no longer gives the exact 95% interval.

The distribution of a Gaussian sum of squares

Definition: Chi-square distribution

For independent $Z_j \sim \mathcal{N}(0,1)$,

$$U = \sum_{j=1}^{\nu} Z_j^2 \sim \chi^2_\nu.$$

This is Gamma with shape $\nu/2$ and rate $1/2$. The number ν is called the degrees of freedom.

For an i.i.d. Gaussian sample,

$$U = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}, \qquad U \text{ is independent of } \bar{X}.$$

Estimating the sample mean leaves $n-1$ independent directions for the centered Gaussian variation.

The Student pivot for an unknown variance

The Student construction from Chapter 2 takes the form

$$\frac{Z}{\sqrt{U/\nu}} \sim t_\nu, \quad Z \sim \mathcal{N}(0,1), \quad U \sim \chi^2_\nu, \quad Z \perp U.$$

Here $\nu = n - 1$, so

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S} \sim t_{n-1}.$$

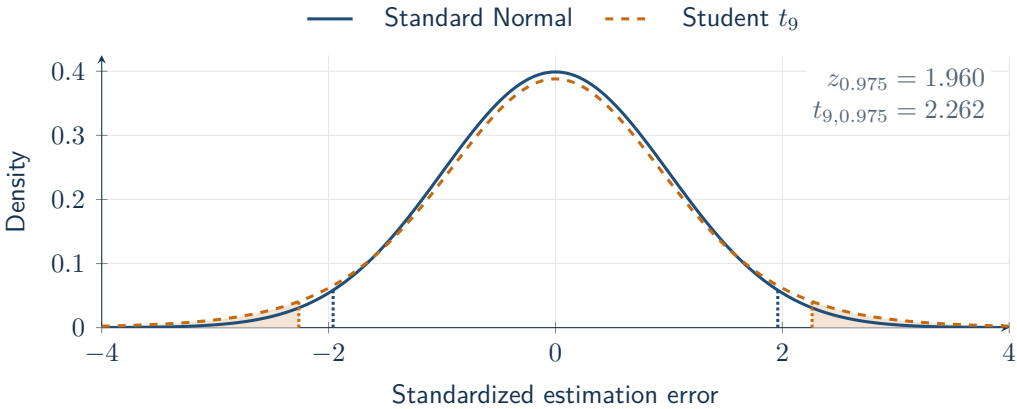
Write $t_{\nu,p}$ for the p-quantile of t_ν.

Result: Gaussian mean, unknown variance

For i.i.d. Gaussian data, an exact $1 - \alpha$ interval is

$$\bar{X} \ \pm \ t_{n-1,\,1-\alpha/2} \frac{S}{\sqrt{n}}$$

Estimating variance makes the tails heavier



The Student law gives more room to large standardized errors. For nine degrees of freedom, the 97.5% quantile is approximately 2.262.

Ten measurements

Suppose the Gaussian model is appropriate and

$$n = 10, \qquad \bar{x} = 10.2, \qquad s = 1.5.$$

The 95% Student interval has margin

$$t_{9,0.975}\frac{s}{\sqrt{n}} \approx 2.262\frac{1.5}{\sqrt{10}} \approx 1.073.$$

Result: 95% confidence interval for the population mean

$$C(x) = [9.13,\ 11.27].$$

Using 1.96 instead gives a smaller margin of 0.930, which ignores the uncertainty in estimating the variance.

Mean intervals beyond Gaussian data

For i.i.d. observations with finite, positive variance, the studentized CLT gives

$$\frac{\sqrt{n}(\bar{X} - \mu)}{S} \xrightarrow{d} \mathcal{N}(0, 1).$$

Thus $\bar{X} \pm 1.96S/\sqrt{n}$ is an asymptotic 95% interval. The same Student formula often gives a similar approximation for large n.

For a finite non-Gaussian sample, the exact t_{n-1} law need not hold.

Skewness and heavy tails can make the approximation poor. There is no universal sample size such as $n = 30$ that guarantees accuracy.

A population mean and a future measurement

The mean interval becomes narrow as we estimate μ more precisely:

$$\bar{X} \pm t_{n-1,\,1-\alpha/2} \frac{S}{\sqrt{n}}.$$

A future observation also has its own measurement noise. Under the same i.i.d. Gaussian model, an exact prediction interval is

$$\bar{X} \pm t_{n-1,\,1-\alpha/2} S \sqrt{1 + \frac{1}{n}}.$$

The prediction interval remains wide even with a very large sample. Individual measurements still vary around the mean.

An exact interval for the exponential rate

For independent exponential waits, their random total satisfies

$$W = \sum_i T_i \sim \text{Gamma}(n, \lambda), \qquad 2\lambda W \sim \chi^2_{2n}.$$

Write $c_{\nu,p}$ for the p-quantile of χ^2_ν.

Result: Exponential rate interval

$$\left[\frac{c_{2n,\alpha/2}}{2W}, \quad \frac{c_{2n,1-\alpha/2}}{2W} \right].$$

For the original five waits, the observed total is $w = 15$ minutes:

$$\widehat{\lambda} = 0.333, \qquad 95\% \text{ interval } [0.108,\ 0.683] \text{ per minute.}$$

Accuracy on independent test examples

For a fixed classifier, let $Y_i = 1$ when its prediction is correct. Under independent examples from one test population,

$$Y_i \sim \text{Bernoulli}(p), \qquad \widehat{p} = \frac{k}{n}, \qquad \text{SE}(\widehat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

The CLT and substitution of $\widehat{p}$ give the Wald interval:

$$\widehat{p} \;\pm\; z_{1-\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

For $k = 84$ correct predictions out of $n = 100$, the approximate 95% interval is $[0.768,\ 0.912]$.

Why the Wald interval can fail

Suppose zero out of twenty tested cases have the outcome of interest. Then $\widehat{p} = 0$, and the estimated standard error is also zero.

$$\text{Wald interval} = [0, 0].$$

But a small positive probability can easily produce twenty zero outcomes:

$$p = 0.05 \quad \Longrightarrow \quad \mathbb{P}(k = 0) = 0.95^{20} \approx 0.358.$$

Substituting a boundary estimate into the variance collapses the interval. Small counts also make Gaussian approximations unreliable.

An interval that retains the candidate variance

For each candidate probability p, use its variance in the standardized error:

$$\frac{\widehat{p} - p}{\sqrt{p(1-p)/n}}.$$

Keep the candidates for which the error is at most $z = z_{1-\alpha/2}$ in magnitude:

$$n(\widehat{p} - p)^2 \leq z^2 p(1-p).$$

Solving this quadratic inequality for p gives the Wilson interval.

This construction also uses a Gaussian approximation. Its coverage is generally much better behaved near the boundaries.

The Wilson proportion interval

Result: Wilson interval, nominal level $1-\alpha$

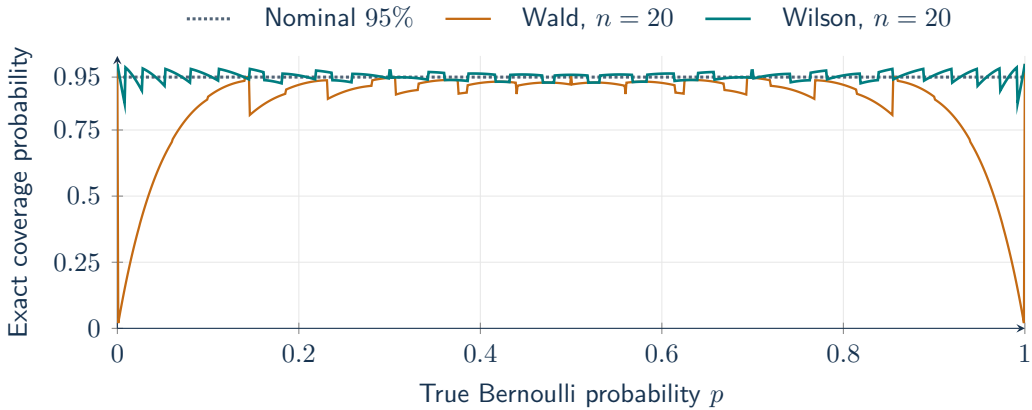
With $z = z_{1-\alpha/2}$, the interval is

$$\frac{\widehat{p} + \dfrac{z^2}{2n} \;\pm\; z\sqrt{\dfrac{\widehat{p}(1-\widehat{p})}{n} + \dfrac{z^2}{4n^2}}}{1 + \dfrac{z^2}{n}}.$$

For zero outcomes in twenty trials, the 95% interval is

$$\left[0,\ \frac{1.96^2}{20 + 1.96^2}\right] \approx [0,\ 0.161].$$

Nominal confidence and actual coverage



For $n = 20$ and each true p, sum binomial probabilities of counts whose interval contains p. Discrete counts make coverage fluctuate.

Using the observed distribution

When a statistic has no convenient sampling formula, the observations can supply an approximate population law.

Definition: Empirical distribution

Put probability $1/n$ on each observed row:

$$F_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{x_i \le t\}}.$$

Repeated values receive the sum of their masses.

Drawing from this law means selecting an observed row uniformly at random.

One bootstrap sample

Start with the five observed waits

$$x = (1, 2, 3, 4, 5), \qquad \bar{x} = 3.$$

Draw five independent indices uniformly from $\{1, \ldots, 5\}$, allowing repeated indices. One possible result is

$$x^* = (1, 1, 2, 4, 5), \qquad \bar{x}^* = 2.6.$$

Other draws omit some observations and repeat others.

The star denotes a random resample, conditional on the fixed observed data. Each bootstrap sample has the original sample size n.

Bootstrap replicates of a fitted statistic

For a statistic $\widehat{\theta} = T(x_1, \ldots, x_n)$:

1. Draw n indices independently and uniformly, with replacement.
2. Form the resampled rows $x_1^*, \ldots, x_n^*$.
3. Recompute the whole statistic, $\widehat{\theta}^* = T(x_1^*, \ldots, x_n^*)$.
4. Repeat to get B replicates $\widehat{\theta}^{*(1)}, \ldots, \widehat{\theta}^{*(B)}$.

The same fitting rule applies to every replicate. The examples here use $B = 20{,}000$ to stabilize the simulated quantiles.

The error distribution we try to approximate

The real estimation error is $\widehat{\theta} - \theta$. In the bootstrap population, the fitted counterpart of θ is $\widehat{\theta}$. The bootstrap error is therefore

$$\widehat{\theta}^* - \widehat{\theta}.$$

For an i.i.d. sample mean with finite, positive population variance, the centered bootstrap error approximates the sampling error as n grows:

$$\mathcal{L}\Big(\sqrt{n}(\bar{X}^* - \bar{x}) \mid x\Big) \;\approx\; \mathcal{L}_{\mu}\Big(\sqrt{n}(\bar{X} - \mu)\Big)\,.$$

Here $\mathcal{L}$ denotes a distribution; conditioning on x fixes the observed data. Other statistics require their own approximation conditions.

A bootstrap standard error

Result: Estimated spread of the statistic

With $\overline{\theta}^* = B^{-1}\sum_b \widehat{\theta}^{*(b)}$,

$$\widehat{\mathrm{SE}}_{\mathrm{boot}} = \sqrt{\frac{1}{B-1}\sum_{b=1}^{B}(\widehat{\theta}^{*(b)} - \overline{\theta}^*)^2}.$$

For the five waits, the exact conditional bootstrap calculation gives

$$\mathbb{E}^*[\bar{X}^*] = 3, \qquad \mathrm{Var}^*(\bar{X}^*) = \frac{2}{5} = 0.4.$$

The usual sample estimate S^2/n is 0.5 here. The empirical distribution uses denominator n in its own variance.

Percentile bootstrap intervals

Let q_p^* denote the empirical p-quantile of the bootstrap estimates.

Definition: Percentile interval

$$[q^*_{\alpha/2},\ q^*_{1-\alpha/2}].$$

Keep the central $1-\alpha$ fraction of the bootstrap estimates.

This is a direct interval construction. Its coverage is approximate and depends on the statistic and sampling distribution.

Inverting centered bootstrap errors

The approximate central error bounds are

$$q^*_{\alpha/2} - \widehat{\theta} \lesssim \widehat{\theta} - \theta \lesssim q^*_{1-\alpha/2} - \widehat{\theta}.$$

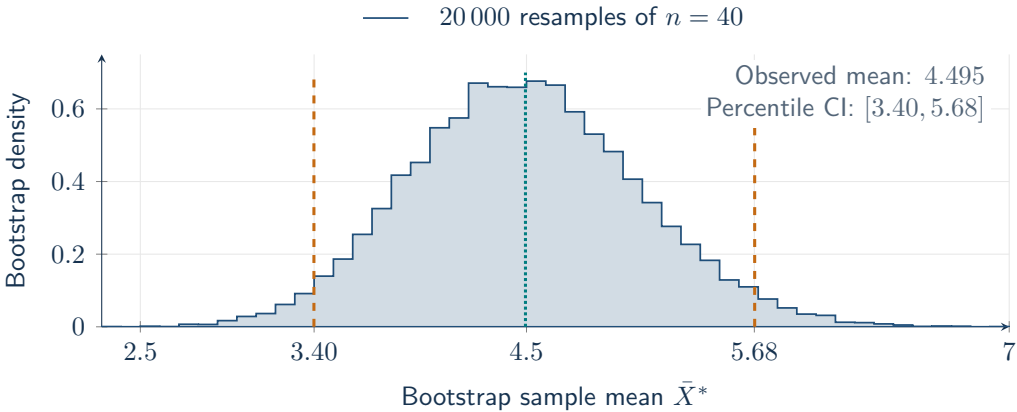
Rearranging for θ gives another construction.

Definition: Basic bootstrap interval

$$[2\widehat{\theta} - q^*_{1-\alpha/2}, \quad 2\widehat{\theta} - q^*_{\alpha/2}].$$

The basic interval reflects the bootstrap quantiles about $\widehat{\theta}$. It can differ from the percentile interval.

Bootstrap estimates of a mean waiting time



Forty illustrative exponential waits have mean 4.495 minutes. The 20,000 bootstrap means give percentile interval $[3.399,\ 5.682]$ minutes.

Two classifiers on the same test examples

Both fixed classifiers use the same 200 independent examples.

	B correct	B incorrect
A correct	150	30
A incorrect	10	10

$$\widehat{p}_A = 0.90, \qquad \widehat{p}_B = 0.80, \qquad \widehat{\Delta} = 0.10.$$

The estimated advantage is ten percentage points. Its uncertainty depends on how the models' successes overlap.

A difference for each example

Let $d_i = \mathbf{1}_{\{A \text{ correct}\}} - \mathbf{1}_{\{B \text{ correct}\}}$. The rows give

$$d_i = \begin{cases} +1 & \text{on 30 examples,} \\ -1 & \text{on 10 examples,} \\ 0 & \text{on 160 examples.} \end{cases}$$

$$\widehat{\Delta} = \bar{d} = 0.10, \qquad \widehat{\mathrm{SE}}(\bar{d}) = \sqrt{\frac{0.20 - 0.10^2}{199}} \approx 0.0309.$$

A Gaussian approximation gives $0.10 \pm 1.96(0.0309)$, or approximately $[0.039,\ 0.161]$.

Bootstrap the paired test rows

Each row contains one example and both classifiers' results:

$$(y_i, \widehat{y}_{A,i}, \widehat{y}_{B,i}).$$

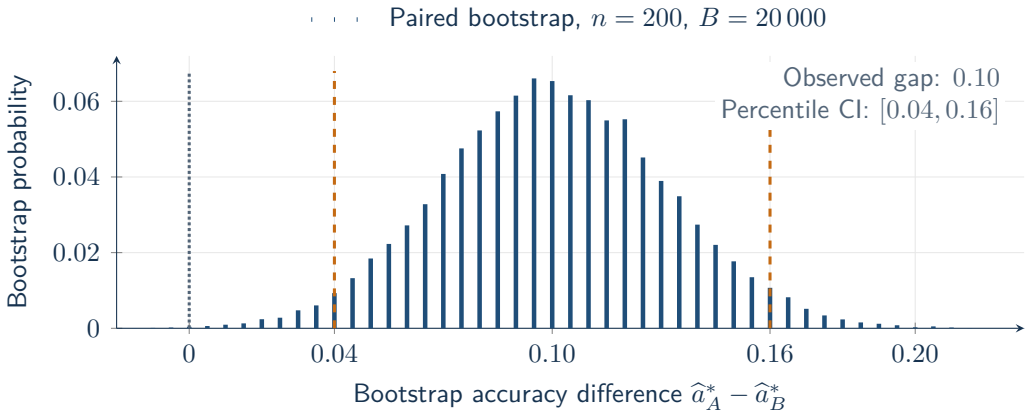
Draw one list of row indices and use it for both models. In each bootstrap sample, recompute

$$\widehat{\Delta}^* = \widehat{p}^*_A - \widehat{p}^*_B.$$

Resampling the models independently destroys their observed association.

Indeed, $\mathrm{Var}(\widehat{p}_A - \widehat{p}_B) = \mathrm{Var}(\widehat{p}_A) + \mathrm{Var}(\widehat{p}_B) - 2\,\mathrm{Cov}(\widehat{p}_A, \widehat{p}_B)$.

Paired bootstrap differences



20,000 paired resamples give percentile interval $[0.04,\ 0.16]$. Both fitted models stay fixed. Only the sampled test examples change.

Metrics built from several counts

Let TP, FP, and FN count true positives, false positives, and false negatives. Then

$$\text{precision} = \frac{TP}{TP + FP}, \qquad \text{recall} = \frac{TP}{TP + FN}, \qquad F_1 = \frac{2TP}{2TP + FP + FN}.$$

Resample complete test rows and recompute all counts, then the metric. For a model comparison, recompute both metrics using the same rows.

If a denominator is zero, specify what the metric means before resampling. Rare positive cases can make the bootstrap distribution unstable.

Averaging per-row precision or per-row F_1 values does not reproduce these ratios of aggregate counts.

More resamples and more observations

The original sample size n controls how much information the data contain. The number of bootstrap replicates B controls simulation accuracy.

$$\text{Larger } B \quad \Longrightarrow \quad \text{more stable simulated standard errors and quantiles.}$$

Once those summaries stabilize, further resampling leaves the statistical uncertainty essentially unchanged.

The resampling unit follows the observation process

Ordinary row resampling assumes that rows act as independent sampling units.

Repeated measurements from one user share user-specific effects. Resampling users together with their measurements preserves that grouping.

Nearby time points can be dependent. Resampling suitable consecutive blocks can preserve some of that dependence.

Independent row resampling can underestimate uncertainty for dependent data.

Observed data can miss possible outcomes

With zero recorded rare events, every ordinary bootstrap sample also has zero. The empirical distribution cannot generate an unseen event type.

For i.i.d. $X_i \sim \mathrm{Uniform}(0, \theta)$, let $M = \max_i X_i$. Then $M < \theta$ almost surely. Every resampled maximum satisfies

$$M^* \leq M, \qquad \mathbb{P}^*(M^* \leq M) = 1.$$

The upper percentile endpoint is at most M, so the percentile interval for θ has coverage zero.

Test uncertainty and training uncertainty

Resampling a test set while holding the model fixed measures uncertainty from which test examples we observed.

To study variation in the fitted model, resample training data and refit the estimator in every replicate.

If model selection is part of the procedure being studied, reproduce that selection too. For the fixed-model accuracy claim in Chapter 4, evaluate on independent test observations that were not used to fit or select the model.

The sample must match the population claim

If the dataset covers only one type of user, resampling it cannot reveal performance for absent user groups.

Sampling bias and distribution shifts require information or assumptions about the missing population. Repeated use of the same observed rows does not supply that information.

An interval statement includes its assumptions

For the ten Gaussian measurements,

$$\widehat{\mu} = 10.2, \qquad 95\% \text{ confidence interval } [9.13, 11.27].$$

This reports uncertainty about the **population mean**, assuming independent measurements from one Gaussian distribution.

The Student construction gives exact coverage under that model. A bootstrap or CLT interval needs its corresponding approximation conditions.